Proving angle relationships

1

a Yes **b** Yes **c** No **d** No

e No

2

a Yes **b** No **c** Yes **d** No

3

a No **b** No **c** Yes **d** Yes

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To prove:

	Statements	Reasons
1.	$m\angle PQS = (5x + 8)^\circ$ $m\angle SQR = (2x - 3)^\circ$ $m\angle PQS = 145^\circ$	Given
2.	$\angle PQS$ and $\angle SQR$ are adjacent angles	Definition of adjacent angles
3.	$m\angle PQS + m\angle SQR = m\angle PQR$	Angle addition postulate
4.	$(5x + 8) + (2x - 3) = 145$	Substitution
5.	$7x + 5 = 145$	Simplification
6.	$7x = 140$	Subtraction property of equality
7.	$x = 20$	Division property of equality

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To prove: $\angle 1 \cong \angle 3$

	Statements	Reasons
1.	$\angle 1$ and $\angle 3$ are vertical angles	Given
2.	$\angle 1$ and $\angle 2$ form a linear pair, $\angle 3$ and $\angle 2$ form a linear pair	Definition of a linear pair
3.	$\angle 1$ and $\angle 2$ are supplementary, $\angle 3$ and $\angle 2$ are supplementary	If two angles form a linear pair, then they are supplementary (linear pair theorem)
4.	$\angle 1 \cong \angle 3$	Angles supplementary to the same angle or to congruent angles are congruent (congruent supplements theorem)

Additional questions

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To prove:

	Statements	Reasons
1.	$2x + 2 = 3x - 10$	Given
2.	$2x + 12 = 3x$	Subtraction property of equality
3.	$12 = x$	Addition property of equality

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To prove: $\angle AXB$ **and** $\angle BXC$ **are supplementary**

	Statements	Reasons
1.	$\angle AXB$ and $\angle BXC$ are a linear pair of angles	Given
2.	$\angle AXC$ is a straight angle	Definition of a linear pair of angles
3.	$m\angle AXC = 180^\circ$	The measure of a straight angle is 180°
4.	$m\angle AXC = m\angle AXB + m\angle BXC$	Angle addition postulate
5.	$m\angle AXB + m\angle BXC = 180^\circ$	Converse of transitive property of equality
6.	$\angle AXB$ and $\angle BXC$ are supplementary	Definition of supplementary angles

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To prove: $\angle AXB \cong \angle CXD$

	Statements	Reasons
1.	$\overline{AC}$ is a straight line segment	Given
2.	$\overline{BD}$ is a straight line segment	Given
3.	$\angle AXB$ and $\angle BXC$ are a linear pair	Definition of a linear pair
4.	$\angle BXC$ and $\angle CXD$ are a linear pair	Definition of a linear pair
5.	$\angle AXB$ and $\angle BXC$ are supplementary	Linear pairs postulate
6.	$\angle BXC$ and $\angle CXD$ are supplementary	Linear pairs postulate
7.	$\angle AXB \cong \angle CXD$	Congruent supplements theorem

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To prove: $\angle A \cong \angle B$

	Statements	Reasons
1.	$\angle A$ and $\angle C$ are complementary	Given
2.	$\angle B$ and $\angle C$ are complementary	Given
3.	$m\angle A + m\angle C = 90^\circ$	Definition of complementary angles
4.	$m\angle B + m\angle C = 90^\circ$	Definition of complementary angles
5.	$m\angle A + m\angle C = m\angle B + m\angle C$	Converse of transitive property of equality
6.	$m\angle A = m\angle B$	Converse of addition property of equality
7.	$\angle A \cong \angle B$	Definition of congruent angles

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To prove: $\angle A$ **and** $\angle B$ **are congruent**

	Statements	Reasons
1.	$\angle A$ and $\angle B$ are right angles	Given
2.	$m\angle A = 90^\circ, m\angle B = 90^\circ$	Definition of right triangles
3.	$m\angle A = m\angle B$	Transitive property of equality
4.	$\angle A \cong \angle B$	Definition of congruent angles

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